知识速查与简答要点

题目怎么问 · 答案必须写到什么 · 可用英文表达

按关键词找到对应条目。中文部分列出回答需要包含的意思；英文段落可以直接用于组织答案。**题目只问定义，就先写定义；问类型、方法、因素或优点，就写完整对应内容。**题目要求比较或举例时，再补相应部分。

题目动词	先写什么	常见遗漏
Define 定义	类别 + 区分特征；例如 temporary + unique。	只写中文译名或泛泛说“很重要”。
List / Identify 列出 / 指认	按题目要求列全或从情景中准确识别。	问六类需求，只列功能和非功能。
Explain 解释	含义或原因；把条件与结论连起来。	写了规则，没有解释为何取最大/最小。
Compare / Distinguish 比较	按相同维度分别写双方。	只解释一方，或用两个无关维度比较。
Calculate / Determine 计算	公式、代入、结果、单位、结论。	只写最后一个数字。
Construct / Draw 构建 / 作图	轴或节点含义、完整条件、标签与结果。	有方格或箭头，缺少依赖或图例。

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先准备老师明确举过的解释内容：Intangible Values；Identifying Cultural Characteristics；Key Factors That Affect Culture Development；Types of Requirements；RAM；Benefits of Gantt Charts。下文已分别给出完整回答。

L1 项目管理基础

知识与术语	答题要点与英文表达
Project 项目 / Operations 运营	**Define a project. Distinguish projects from operations.** **Temporary 临时性 + Unique 独特性。**有明确开始和结束，创造独特产品、服务或成果。运营持续、重复。临时不等于必须很短，项目成果也可长期使用。 A project is a **temporary endeavor** undertaken to create a **unique product, service, or result**. Operations are ongoing, repetitive activities that sustain the organization. 开发新的预约系统是项目；系统上线后每天处理预约是运营。Program 项目集是一组为共同目标而协调管理的相关项目。
Business Value 商业价值 / Intangible Values 无形价值	**Define intangible values and give examples.** **老师明确举例。**先写 non-physical benefits，再说明项目为干系人创造收益。无形例子：Goodwill 商誉、Brand recognition 品牌认知、Public benefit 公共利益、Trademarks 商标、Strategic alignment 战略一致性、Reputation 声誉。 **Intangible values are non-physical benefits** that a project creates for its stakeholders, such as goodwill, brand recognition, public benefit, trademarks, strategic alignment, and reputation. 本课有形要素：Monetary assets 货币资产、Stockholder equity 股东权益、Utility 效用、Fixtures 固定装置、Tools 工具、Market share 市场份额。不要把“难量化”写成“没有价值”。

知识与术语	答题要点与英文表达
Triple Constraint 三重约束	**Explain the triple constraint.** **Scope 范围、Time 时间、Cost 成本**三项要列全；说明相互关联。Quality 质量位于三角形中央，改变一项后需要评估其他方面。 The triple constraint consists of **scope, time and cost**. These constraints are interrelated, so a change in one requires the project manager to assess its effects on the others and on quality. 增加网上付款功能会增加工作范围，经理需要重新评估进度、费用与质量；不能直接断言三者必定同时增加。
Process Groups 过程组 / Life Cycle 生命周期	**List the process groups. Explain the life-cycle phases.** **五个过程组：**Initiating 启动、Planning 规划、Executing 执行、Monitoring and Controlling 监控、Closing 收尾。**四个生命周期阶段：**Defining 定义、Planning 规划、Executing 执行、Closing 收尾。过程组可以在阶段内重复，不是一一对应。 **Initiating** defines and authorizes the project or phase. **Planning** establishes scope, objectives and the course of action. **Executing** performs the planned work. **Monitoring and Controlling** tracks performance and makes adjustments. **Closing** formally completes the project, phase or contract. The four life-cycle phases are Defining, Planning, Executing and Closing. 读生命周期图：横轴 Time，纵轴 Level of effort；执行阶段投入最高，相邻阶段可以交叠。投入低不等于阶段不重要。
PM / Agile / Socio-Technical	**Explain the project manager's role, Agile, or the socio-technical approach.** 项目经理负责整体绩效并协调资源与人员。**Agile 敏捷：incremental 增量、iterative 迭代、客户合作、适应变化。**本课小团队 **4-8 人**，每轮 **1-4 周。**社会技术方法要同时写技术面与社会文化面。 The project manager directs and coordinates the project team and is responsible for overall project performance. Agile uses an **incremental and iterative** approach, customer collaboration and adaptation to change. The **socio-technical approach** combines technical planning, scheduling and control with leadership, teamwork, negotiation and customer expectations. 增加预约功能是增量；按患者反馈改善预约页面是迭代。算工期是技术面；协商医院安装时段是社会文化面。Rolling wave：近期工作细化，较远工作随信息增加再细化，每轮回顾调整。
Drivers 管理需求 / Maturity 成熟度	**Explain the drivers of project management and project management maturity.** **五个驱动：**产品生命周期缩短、知识增长、三重底线、更重视客户、小项目累积产生大问题。Triple bottom line 为 **Planet 环境、People 人与社会、Profit 利润。**成熟度看组织的方法是否系统、一致并持续改进。 Demand for project management is driven by shorter product life cycles, knowledge growth, the triple bottom line, increased customer focus, and the cumulative demands of small projects. **Project management maturity** describes how consistently an organization applies and improves its practices. Maturity models support benchmarking and process improvement. 低成熟度：临时做法多；中等：有方法、培训和支持；高：制度化且持续改进。不能仅凭“科研”或“高科技”判断成熟度。
Ten Knowledge Areas 十个知识领域	**List the knowledge areas or identify the area involved.** 按“管理什么”分类，沿用本课的 Time、Human Resource 名称。十项：**Integration、Scope、Time、Cost、Quality、Human Resource、Communications、Risk、Procurement、Stakeholder。** The ten knowledge areas are **Integration, Scope, Time, Cost, Quality, Human Resource, Communications, Risk, Procurement and Stakeholder Management**. Knowledge areas classify what is managed; process groups classify management processes by their purposes. 整合协调变更；范围管工作边界；时间管进度；成本管费用；质量管满足要求；人力管人员；沟通管信息；风险管不确定性；采购管外部取得；干系人管相关方需求与参与。

L2 环境、干系人与组织

知识与术语	答题要点与英文表达
Greenfield / Brownfield / SITES	**Distinguish greenfield and brownfield projects.** **Greenfield** 在全新或未开发场地开展；**Brownfield** 在既有场地、设施上改造或扩建。判断依据是已有条件，既有场地不必然有污染，全新场地也仍受法规和预算约束。 A **greenfield project** develops a vacant site without constraints imposed by prior work on that site. A **brownfield project** expands or revamps an existing facility and must account for its current condition. 空地建医院与改造旧医院分别对应两类。SITES：Social 社会、Information 信息、Technology 技术、Economic 经济、Sustainability 可持续性。
TOWS Matrix 战略矩阵	**Construct or explain a TOWS matrix.** 内部 **S Strengths 优势、W Weaknesses 劣势**；外部 **O Opportunities 机会、T Threats 威胁**。四格必须是两类条件的配对行动，不能只写四个词。 A TOWS matrix matches internal strengths and weaknesses with external opportunities and threats. **SO** uses strengths to exploit opportunities; **ST** uses strengths to address threats; **WO** uses opportunities to reduce weaknesses; **WT** reduces weaknesses and avoids threats. 自身数据分析强=S，英文写作弱=W；有奖学金=O，申请竞争强=T。SO：用分析项目申请奖学金；ST：用项目成果突出竞争优势；WO：为申请奖学金补写作；WT：改进写作并准备适合能力的多个选项。
Stakeholder 干系人 / Power-Interest	**Define a stakeholder and use the power-interest matrix.** 能影响、受影响或认为自己受影响的人、群体、组织，**反对者也属于干系人**。内外按相对项目组织的位置判断。矩阵轴是 Power 权力、Interest 关注程度。 A stakeholder is an individual, group or organization that can affect, be affected by, or perceive itself to be affected by project decisions, activities or outcomes. In the power-interest matrix, use **Manage closely** for high/high, **Keep satisfied** for high power/low interest, **Keep informed** for low power/high interest, and **Monitor** for low/low. 项目经理与团队通常属于内部；供应商、监管机构等通常属于外部。题目若给定了不同组织边界，以该边界判断。
Cooperation-Threat Matrix 合作与威胁矩阵	**Classify stakeholders and select the strategy.** 两个轴分别判断。低合作低威胁：**Marginal / Monitor**；低合作高威胁：**Non-supportive / Defend**；高合作低威胁：**Supportive / Involve**；高合作高威胁：**Mixed blessing / Collaborate**。 The cooperation-threat matrix assesses stakeholders by their potential for cooperation and threat. **Monitor** marginal stakeholders, **Defend** against non-supportive stakeholders, **Involve** supportive stakeholders, and **Collaborate** with mixed-blessing stakeholders. 愿意协调资源、又有权否决预算的主管可以同时高合作、高威胁。不能把“愿意支持”直接等同于低威胁。
Functional / Projectized / Matrix 组织结构	**Identify and compare organizational structures.** 职能型按专业部门组织，职能经理权力大；项目型按项目组织，项目经理权力大；矩阵型结合两者，可能双重汇报。**Weak 职能经理较强；Balanced 权力较均衡；Strong 项目经理较强。** A **functional structure** groups people by specialized functions. A **projectized structure** organizes people around projects and gives the project manager substantial authority. A **matrix structure** combines both, with shared authority and possible dual reporting. Authority shifts from the functional manager toward the project manager as the matrix becomes stronger. 比较时写同一维度：谁安排工作、谁控制资源、成员向谁汇报。职能型便于专业发展，跨部门协调较难；项目型集中投入但可能重复配置；矩阵共享资源但会发生权力冲突。

知识与术语	答题要点与英文表达
PMI / PMP / PMO / Talent Triangle	**Explain a PMO or distinguish the three PMO types.** **PMI** 专业协会，**PMP** 个人认证，**PMO** 组织内部单位。PMO：Supportive 提供模板建议；Controlling 要求遵守标准；Directive 直接管理项目。 A **PMO** is an organizational unit that supports or governs project management. A supportive PMO provides advice and resources; a controlling PMO requires compliance with standards; a directive PMO directly manages projects. The Talent Triangle includes **Ways of Working, Power Skills and Business Acumen**. 人才三角依次对应管理方法、人际与领导能力、理解业务目标和价值；Power Skills 不是职位权力。
Organizational Culture 文化与冲突	**Define culture; explain how to identify it or what shapes it.** **老师明确举例：识别方法与形成因素要分开。**文化是共享的 norms、beliefs、values、assumptions。识别四法：看实体环境、读组织资料、观察互动、理解故事。形成七因素：技术、环境、地理、奖励、规则流程、关键成员、重大事件。 Organizational culture consists of **shared norms, beliefs, values and assumptions** that guide behavior. Identify it by studying physical characteristics, reading about the organization, observing interactions, and interpreting stories and folklore. Culture development is affected by **technology, environment, geographical location, reward systems, rules and procedures, key organizational members, and critical incidents**. 冲突五法：Avoidance 回避；Accommodation 迁就对方；Competition 争取己方目标；Compromise 双方让步；Collaboration 合作满足主要需求。妥协不是双方完全满足，合作也不是简单停止争论。

L3 范围与责任

知识与术语	答题要点与英文表达
Product Scope / Project Scope 产品与项目范围	**Distinguish product scope and project scope.** **Product scope** 是成果的功能与特征；**Project scope** 是交付这些功能与特征所需的工作。先确定范围，再分解、估算、排程。 Product scope describes the **features and functions** of the product, service or result. Project scope is the **work required to deliver** it with those specified features and functions. Scope management defines and controls what is and is not included in the project. “能够查询、预约、取消”是产品范围；“完成设计、开发、测试、培训”是项目范围。
Types of Requirements 六类需求	**List and explain the types of requirements.** **老师明确举例。**六类：Business、Stakeholder、Solution、Transition and readiness、Project、Quality。Solution 再分 Functional 功能与 Nonfunctional 非功能。 **Business** requirements explain why the project is undertaken. **Stakeholder** requirements express specific stakeholders' needs. **Solution** requirements specify functional capabilities and nonfunctional attributes. **Transition and readiness** requirements support the move to the future state. **Project** requirements specify conditions the project must meet. **Quality** requirements define criteria for successful completion and compliance. 减少等待=业务；护士要查看名单=干系人；能取消预约=功能；指定负载下 2 秒响应=非功能；数据迁移与培训=过渡就绪。只回答功能/非功能会漏四类和层级关系。

知识与术语	答题要点与英文表达
Scope Statement 范围说明书	**Explain what a scope statement contains.** 六项：**Product scope description、Acceptance criteria、Deliverables、Exclusions、Constraints、Assumptions。**交付什么、怎样验收、不做什么、有什么限制与假设，要分别写。 A project scope statement includes the **product scope description, acceptance criteria, deliverables, exclusions, constraints and assumptions**. Deliverables are the results to provide; acceptance criteria are the conditions for accepting them. Exclusions identify work outside the scope, constraints set limits, and assumptions are conditions treated as true for planning. 系统是成果；通过约定测试是验收标准；本期不做支付是排除项；限期上线是约束；医院按期提供测试数据是假设。Control Quality 检查成果合格；Validate Scope 正式接受已完成成果。
WBS / Decomposition / Work Package	**Define WBS, decomposition and a work package.** **WBS** 层级分解项目总范围；**Decomposition** 是分解方法；**Work package** 是最低层。分解过程：识别成果与工作，组织层级，向下细分，分配编号，检查是否充分。100% 规则：不漏不重，子项覆盖父项全部工作。 A **WBS** is a hierarchical decomposition of the total project scope into smaller, manageable components. **Decomposition** divides scope and deliverables into those components. A **work package** is the lowest-level WBS component that can be estimated, assigned, scheduled and monitored. The **100% rule** requires complete coverage without duplication. 工作包不必限定一人或几天。包括规划与收尾；避免一个父项只有一个子项的多余层级；低层活动名称以动词说明工作。Scope baseline=范围说明书+WBS+WBS dictionary。
OBS / RAM 责任分配	**Define OBS or explain RAM with a simple table.** **老师明确举例：RAM。**OBS 按组织层级分；RAM 把任务与人员对应，交叉格表示角色。**任务为行、人员为列，先写清图例。**RAM 又称 Linear Responsibility Chart。 An **OBS** is a hierarchical representation of the organizational units responsible for project work. A **RAM**, also called a linear responsibility chart, links project tasks to team members and shows their roles. Tasks form the rows, people form the columns, and the legend specifies responsibility, support, notification and approval. 图例：○ Responsible 负责，☆ Support 支持，■ Notification 通知，□ Approval 批准。 “开发预约功能”可填工程师○、护士☆、经理■。RACI 是另一套字母，按题给图例回答。

L4 进度与概率

知识与术语	答题要点与英文表达
Activity / Duration / Predecessor	**Read an activity table.** **Activity** 是具体工作；**Duration** 是持续多久；**Immediate predecessor** 是直接依赖的前项。表格至少读清编号、工作、工期、紧前活动。 An activity is a distinct piece of project work. Its duration is the time required to complete it. An immediate predecessor directly precedes another activity according to the dependency relationship. 安装电脑是活动，2 天是工期。在 A-B-D 流程中，D 的紧前是 B，A 是更早的前置。紧前有 B、C 就要两者条件都满足。
AON / Merge / Burst	**Draw or explain an AON network.** **节点=活动；箭头=依赖。**Merge 汇合活动有多个紧前；Burst 分叉活动有多个紧后。每项活动一个节点，按活动表逐行连紧前箭头。 In an **Activity-on-Node network**, nodes represent activities and arrows represent dependencies. A **merge activity** has more than one immediate predecessor, while a **burst activity** has more than one immediate successor. D 等 B、C，就画 B→D、C→D；H 等 E、F、G，就保留三条进入 H 的箭头。节点在纸上的位置不是实际日期。

知识与术语	答题要点与英文表达
ES / EF / LS / LF / SL 九宫格	**Calculate the activity times and slack.** 九宫格：上行 **ES / ID / EF**；中行左 **SL**，右边合并写活动名称；下行 **LS / DUR / LF**。正推取紧前 EF 最大值；逆推取紧后 LS 最小值。 **ES = max(EF of immediate predecessors); EF = ES + duration.** **LF = min(LS of immediate successors); LS = LF - duration.** **SL = LS - ES = LF - EF.** A merge must wait for all predecessors; an activity must finish in time for every successor. 没有紧前的活动从 0 开始。多个最终活动逆推时用同一个项目完工目标。D 等 EF=10、11 的两项，ES=11；B 后面 LS=11、22，LF=11。
Path / Critical Path 路径与关键路径	**List all paths and identify the critical path.** 沿箭头从开始到结束连续枚举，每条路径把活动工期相加；**工期最长的所有路径**都要列为关键路径。 A path is a continuous sequence of activities through the network. The **critical path is the longest-duration path** and determines the minimum project duration under the given dependencies and durations. More than one critical path may exist. Delta 四条路径：A-B-D-F-H=29，A-B-E-H=18，A-C-D-F-H=30，A-C-G-H=22 周；关键路径 A-C-D-F-H，30 周。本题逆推目标为最早完工，关键活动 SL=0。
Total Slack / Free Slack 总时差与自由时差	**Explain slack and judge the effect of a delay.** **Total** 看项目完工；**Free** 看紧后活动最早开始。只有单项工期增加 Δ、其余条件不变时，项目延长量 =max(0,Δ-原总时差)。多项变化要重新排程。 **Total slack** is the time an activity can be delayed without delaying project completion. **Free slack** is the time it can be delayed without delaying the earliest start of any immediate successor. A delay can change successor schedules even when it does not delay the project. Delta 中 B 总时差 1、自由时差 0。B 多用 1 周会让 E 晚开始，项目仍 30 周完成。若 B+1、E+12 同时发生，项目变 31 周；时差不能当作互不影响的额度。
PERT 三点估计	**Calculate the expected duration or reverse-solve an estimate.** **a 乐观、m 最可能、b 悲观。**TE=(a+4m+b)/6；活动标准差 σ=(b-a)/6。反求：b=6TE-a-4m；m=(6TE-a-b)/4；a=6TE-4m-b。 PERT uses optimistic, most likely and pessimistic estimates. **TE = (a + 4m + b) / 6**, and the activity standard deviation is **(b - a) / 6**. The most likely estimate has a weight of four. a=3、m=4、b=8 天，TE=4.5 天、σ=0.8333 天。反求后检查 a≤m≤b；标准差与时间同单位，方差为时间平方。
Z-score / Due Date 标准分数与期限	**Calculate Z, a completion probability, or the required due date.** **Z=(D-μ)/σp；D=μ+Zσp。**μ 为期望完工时间，D 为期限，σp 为题给项目标准差。Z 是相差多少个标准差，无单位；概率用题给对应值。 A Z-score measures the difference between a deadline and the expected completion time in units of the project standard deviation. **Z = (D - μ) / σp**. The required due date is **D = μ + Zσp**; additional time is **Zσp**. 30 周、σp=2.78、期限 32：Z≈0.72；题给累计概率 0.7642，则按时概率 76.42%，超期 23.58%。给约 95% 对应 Z=1.65：期限 34.587≈34.59 周，额外 4.587 周。

L5 进度关系与成本

知识与术语	答题要点与英文表达
Gantt Chart 甘特图	**Read a Gantt chart or explain its benefits.** 横条起点/终点为开始/完成；长度为工期；依赖看箭头或文字。**老师明确举例的五项优点：**易读、呈现网络与基准、便于更新控制、安排资源、易制作。 A Gantt chart is a time-based bar chart showing activity start, finish and duration. Gantt charts are **easy to read**, identify the network together with its **schedule baseline**, allow **updating and project control**, help identify and assign **resources**, and are **easy to create**. 横条从 5 到 8，工期为 3。相邻不证明依赖；并排不证明可以交换。只有计划条不能判断实际落后。读到依赖后可转换成简化 AON，再比较路径。
FS / SS / FF / SF 与 Lag	**State the relationship using cannot … until … .** A 前项、B 后项。**FS**：A 完成，B 才能开始；**SS**：A 开始，B 才能开始；**FF**：A 完成，B 才能完成；**SF**：A 开始，B 才能完成。Lag 是关系中的规定等待时间。 **FS:** B cannot start until A finishes. **SS:** B cannot start until A starts. **FF:** B cannot finish until A finishes. **SF:** B cannot finish until A starts. 同样是开发原型、测试，限制测试开始为 FS，限制测试结束为 FF。旧系统停运要等新系统启用为 SF。若 A 在 10 完成、FS+4、B 工期 2，B 排在 14-16；B 工期仍为 2。
Cost Categories 四组成本分类	**Distinguish the cost categories.** **Direct / Indirect** 按能否直接归属；**Recurring / Nonrecurring** 按是否反复发生；**Fixed / Variable** 按是否随用量变化；**Normal / Expedited** 按正常或加速条件。四组是独立维度。 Direct costs are traceable to specific project work; indirect costs support the project more generally. Recurring costs occur repeatedly; nonrecurring costs occur once or infrequently. Fixed costs remain unchanged within the relevant range; variable costs change with usage. Expedited costs arise when work is accelerated. 活动材料通常直接归属；共用行政费用是间接；持续工资为经常性；初始培训可为一次性；加急运送为加速费用。同一笔费用可同时属于几组分类。
Fully Loaded Rate 综合人工费率	**Calculate the loaded rate and total labor cost.** **综合时薪=基本时薪×(1+附加比例)**；总费用=综合时薪×工时。附加 65% 就乘 1.65，0.65 只计算附加部分。 The fully loaded labor rate includes base pay and additional employment costs. **Loaded hourly rate = base hourly rate × loading factor.** Total labor cost equals the loaded hourly rate multiplied by hours worked. $37.50/hour×1.65=$61.875/hour；再乘 80 小时=$4,950。费率单位为金额/小时，总成本单位为金额；最后按题目要求舍入。
Time-phased Budget 时间分段预算	**Explain or read a time-phased budget.** 把计划费用放到预期发生的时间段。**本期费用**是某一时间段的支出；**累计费用**是截至该期的总和。本次保留概念与读表。 A time-phased budget allocates planned project costs to the periods in which they are expected to occur. It shows periodic funding requirements and cumulative planned spending. 第 1-3 周分别为 $100、$200、$150；第 3 周本期是 $150，累计是 $450；相邻两期累计值之差等于该期费用。

知识与术语	答题要点与英文表达
Crashing 赶工 / Cost Slope 费用斜率	**Calculate the cost slope and decide whether crashing helps.** **斜率=(赶工费用-正常费用)/(正常工期-赶工工期)。**先看是否影响当前关键路径，再比较单位时间增加费用。Fast-tracking 通过重叠工作，Crashing 通过增加资源或费用。 Crashing adds resources or cost to shorten selected activities. **Cost slope = (crash cost - normal cost) / (normal duration - crash duration).** The selected reduction must affect the controlling critical path to shorten the project. 5 天 $1,000 改为 3 天 $1,600：斜率 $300/day，最多压 2 天。两条关键路径时只压一支未必有用；先检查共有活动或两条路径的共同约束。
AON / AOA / Dummy 比较	**Distinguish AON and AOA; define a dummy activity.** **AON 节点=活动；AOA 箭头=活动，节点=事件。**Dummy 工期为 0、不耗资源，通常用虚线保留依赖。虚活动可以在路径上，工期加 0。 In AON, nodes represent activities. In AOA, arrows represent activities and nodes represent events. A **dummy activity** is a zero-duration activity used in AOA to preserve the correct dependency logic. It consumes no resources and is normally shown as a dashed arrow. 判断“是否需要虚活动”要查真实依赖，而不是数图里有几个交叉点。事件编号可以不同，表达的依赖必须正确。
AOA 10.15 虚活动判断	**Explain why the dummy activities are needed.** D 等 B、C；E 只等 B；G 只等 C。**保留 B、C 两个完成事件，单独设 D 的开始事件**，用两条虚线把 B、C 的完成条件接入 D。 D must wait for both B and C, while E depends only on B and G depends only on C. Separate finish events for B and C, together with two dummy activities leading to D, preserve these different dependencies. 可用 B:2→3、C:2→4，虚线 3→5、4→5，D:5→6；E 从 3 出发，G 从 4 出发。把 B、C 终点直接合并会错误地让 E 等 C、G 等 B。

L6 风险管理

知识与术语	答题要点与英文表达
Risk 风险 / Four Stages 四阶段	**Define project risk and list the risk management stages.** **Uncertain event or condition**；发生后影响项目目标，可正可负。四阶段为识别、概率后果分析、缓解策略、控制与文档化。 A project risk is an **uncertain event or condition** that, if it occurs, has a positive or negative effect on project objectives. The four stages are **risk identification; analysis of probability and consequences; risk mitigation strategies; and control and documentation**. 头脑风暴列未来可能的问题=Identification。已经确定发生的故障是当前问题，不再只是尚未发生的可能性。
RBS 风险分解结构	**Define RBS and classify risk sources.** 按**风险来源**作层级分类，帮助逐类检查。五类来源：**Financial、Technical、Commercial、Execution、Contractual/legal。**与按工作分解的 WBS 区分。 A Risk Breakdown Structure is a **hierarchical grouping of potential sources of project risk**. It helps the team identify risks systematically across financial, technical, commercial, execution, and contractual or legal categories. 资金不到位=财务；接口不兼容=技术；竞争影响需求=商业；关键人员缺勤=执行；合同责任争议=合同法律。识别可用头脑风暴、专家意见、历史资料、团队评估。

知识与术语	答题要点与英文表达
Risk Impact Matrix 风险影响矩阵	**Place risks in a likelihood-consequence matrix.** 两轴分别为 **Likelihood 可能性**和 **Consequence 后果**。先读轴方向，再逐项放格子；相同可能性下，后果越严重越需关注，反之同理。 A risk impact matrix positions risks according to their likelihood and consequences. It helps compare priorities and directs attention to risks with greater likelihood or more serious consequences. 纵轴低到高、横轴低到高：低可能性高后果放右下；中可能性高后果放右中；高高放右上。定性格子不自动对应某个 RF 数值，数值等级按评分规则另判。
Risk Responses 风险应对	**Identify or explain the risk response.** **Accept** 已知后承担；**Minimize / Mitigate** 降低可能性或后果；**Share** 共同承担；**Transfer** 把指定后果交第三方。Contingency reserves 事先预留余量；Workaround 是事后临时应对。 Acceptance consciously retains a risk. Minimization reduces its probability or consequences. Sharing involves joint responsibility for risks and potential benefits. Transfer shifts responsibility for specified consequences to a third party. Contingency reserves provide time or funds for risk-related needs. 协议未定仍继续计划=Accept；与另一家公司共同开发=Share；买保险=Transfer；加强检查=Minimize。合作不自动等于转移，接受不等于没识别。
Pf / Cf / RF 风险评分	**Calculate and classify the project risk factor.** **两组分别平均。**Pf 为概率评分平均；Cf 为后果评分平均；RF=Pf+Cf-PfCf。**Low <0.30；Medium 包含 0.30-0.70；High >0.70。** Calculate Pf as the mean of the probability scores and Cf as the mean of the consequence scores. Then use **RF = Pf + Cf - PfCf**. Classify the project as low, medium or high risk using the stated thresholds. 给 5 项概率评分就除以 5；后果组有 4 项则除以 4。空白不计入，明确的 0 要计入。RF 不是失败概率百分比；用未舍入值先判级。
Risk Scoring 完整计算答案	**Show the calculation and give the risk classification.** 五项概率 0.7、0.7、0.5、0.5、0.3；四项后果 0.9、0.7、0.3、0.5。先算 Pf=0.54、Cf=0.60，再写 RF=0.816 与 High risk。 Pf = (0.7 + 0.7 + 0.5 + 0.5 + 0.3) / 5 = 0.54. Cf = (0.9 + 0.7 + 0.3 + 0.5) / 4 = 0.60. **RF = 0.54 + 0.60 - (0.54 × 0.60) = 0.816.** Since RF is greater than 0.70, the project is classified as **high risk**. 若 Pf=0.50、Cf=0.40，RF=0.70，Medium。若 RF=0.704，则 High，即使保留两位小数会显示 0.70。

五种矩阵与责任表：按轴查答案

TOWS：内部条件与外部条件配对

内部 ↓ / 外部 →	Opportunity 机会	Threat 威胁
Strength 优势	SO：用优势争取机会	ST：用优势应对威胁
Weakness 劣势	WO：借机会弥补劣势	WT：减少劣势与威胁

写策略要用到题干中的两个具体条件。例如数据分析强 + 有研究奖学金 → 用分析项目准备研究申请。只有“努力改进”没有说明配对理由。

Power-Interest：权力与关注程度

Power 权力 ↓ / Interest 关注 →	Low 低	High 高
High 高	Keep satisfied 保持满意	Manage closely 密切管理
Low 低	Monitor 关注动态	Keep informed 保持知情

Cooperation-Threat：合作潜力与威胁潜力

Cooperation 合作 ↓ / Threat 威胁 →	Low 低	High 高
High 高	Supportive 支持型 Involve 邀请参与	Mixed blessing 利弊并存型 Collaborate 深入合作
Low 低	Marginal 边缘型 Monitor 关注动态	Non-supportive 不支持型 Defend 防范威胁

两轴独立判断。能提供很大帮助的主管，也可能有权阻止项目，所以会落在高合作、高威胁。

RAM：任务与人员对应

任务 / 人员	项目经理	工程师	护士代表
确定规则	□	☆	○
开发功能	■	○	☆
验收功能	□	☆	○

○ Responsible；☆ Support；■ Notification；□ Approval。读“开发功能”一行：工程师负责，护士支持，经理接收通知。图例随题目给定，先读图例再解释。

Risk Impact Matrix：可能性与后果

Likelihood ↓ / Consequence →	Low 低	Medium 中	High 高
High 高			D
Medium 中			B
Low 低		C	A

A：低可能性、高后果；B：中可能性、高后果；C：低可能性、中后果；D：高可能性、高后果。D 最需关注；A 与 C 同为低可能性，A 后果更严重。按给定分区定等级，不从字母或颜色猜 RF。

答案写完后，按题目再核对一次

定义有没有写到区别于其他概念的特征；列举有没有漏项；比较有没有用同一维度；计算有没有保留单位与结论；图表有没有完整表达题目的条件。英文答句中的项目名、活动编号和数值，换成当前题目的条件。